Name: _____ Department: _____ Student ID: _____

Team No.: _____ Date: _____ Lecturer's Signature: _____

Starting Time: _____ End Time: _____

Student preparations

- *Principles of Physics, 10th ed. Walker, Halliday and Resnick, Ch. 25 Interference; Ch. 26 Diffraction*
- *PASCO manuals: WA-9896 Ripple Generator and Light Source; WA-9899 Ripple Tank System*

Goals

- We will observe how speed and/or direction of a wave change when it propagates in a medium that is not uniform in physical properties.
- We will observe how two waves interfere each other when they meet at the same medium.
- We will also try to understand how the frequency of wave changes depending on the speed of the source.

Theoretical Background

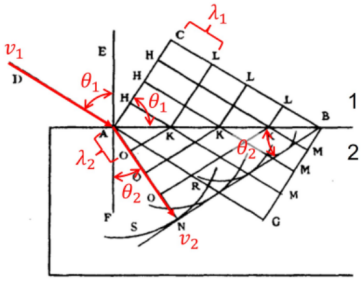
A wave is a propagating dynamic disturbance of one or more quantities in a continuous medium (or continuum). While a wave typically propagates (or travels) along a straight line, it may change its direction and/or speed if the space, or a medium, where it travels is not uniform in properties. Since the frequency f of a wave does not change depending on the medium but remains constant, the change in wave speed $v = f\lambda$ is equivalent to the change in wavelength λ. In the following, we investigate important cases of a wave changing its direction of propagation.

- **Refraction**

  Refraction is the change in direction of a wave passing from one medium to another, or from a gradual change in the medium, which is ascribed to the change in wave speed v and wavelength λ. Refraction in general may occur even without changes in direction, for instance, if the wave enters another medium perpendicularly to the interface.

  Let us consider a wave propagating from medium 1 to medium 2 across an interface as depicted in Fig. 1. In medium 1, it propagates at the speed $v_1 = f\lambda_1$ and is incident on the interface at an angle θ_1 which is measured off the direction normal to the interface. In medium 2, the wave leaves the interface at an angle θ_2 and propagates at the speed $v_2 = f\lambda_2$. Since all wavefronts (for instance lines of L-K-O in Fig. 1) must be continuous across the interface, the distance between two successive wavefronts measured on the interface is given as $L = \lambda_1/\sin\theta_1 = \lambda_2/\sin\theta_2$. This equation can be rewritten using the wave speed as follows, which is known as the Snell's Law of Refraction.

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2} \tag{1}$$



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Figure 1: Christian Huygens's illustration of wave refraction.

- **Diffraction & Interference**
  Diffraction is defined as the bending of waves around the corners of an obstacle or through an aperture into the region of geometrical shadow of the obstacle. This phenomenon is due to the fact that the medium of wave propagation is a continuum and thereby a disturbance on any point of the medium will be transferred to its surroundings. It is described by the Huygens–Fresnel principle stating that every point on a wavefront is itself the source of spherical wavelets, and the secondary wavelets emanating from different points mutually interfere. As a result, the wave displacement at any subsequent point appears as the superposition (or mathematical sum) of these secondary waves.

  The apparent angle of bending, or diffraction, is determined as a function of geometry of the aperture and wavelength. In general, the angle will become larger for smaller aperture width and larger wavelength. It is because the maximum angle accounts for the complete destructive interference of all wavelets passing the aperture. The simplest case is given as a plane wave of λ passing a slit of a finite width a perpendicularly. For a complete destructive interference to occur, a wavelet passing a spot on one side of the aperture must cancel with another passing a spot $a/2$ away and on the other side. Such condition demands that the two rays interfering with each other have a phase difference of π or a path difference of $\lambda/2$. If we assume that the two rays arrive at a place far away from the aperture, the path difference becomes approximately equal to $\frac{a}{2}\sin\theta$. Finally, the condition for the complete destructive interference is found to be $\lambda = a\sin\theta$.

- **Double Slit Interference**

  The understanding of the constructive interference becomes simpler when we consider only two wave rays simultaneously passing through two slits separated by the distance d (see Figure 2). Since half circular (or half spherical) wavelets will emerge from the other side of the infinitesimally narrow slits, the double slit acts like two point sources. The positions of maximum intensity, or the constructive interference, are found when the path difference is λ or its integer multiples. Since the path difference will be $d\sin\theta$ for the rays tilted at an angle θ away from the original direction of propagation, the condition of the constructive interference is given as follows.

$$d\sin\theta = m\lambda \qquad (m = \text{integers}) \tag{2}$$

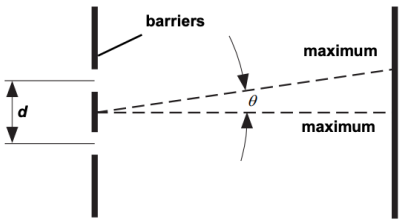


Figure 2: The schematic setup of the double slit interference

- **Doppler Effect**

  Whereas frequencies of a wave will not typically change during reflection, refraction, or diffraction in a continuum, they may change if the source and/or the receiver moves relatively to the medium by a speed lower than the wave speed. The change in frequency is described by the Doppler effect as follows. Suppose that a source emits a wave at a frequency f_0, which subsequently travels in a medium at the speed $v_0 = f_0\lambda_0$. If the source also moves at a speed v_s towards(away from) the receiver, the latter will observe the wavelength to be shorter(longer) as $\lambda = v_0/f = \lambda_0 - v_s/f_0 (= \lambda_0 + v_s/f_0)$. Instead, if the receiver itself moves at a speed v_r towards(away from) the source, the receiver will observe the frequency to be higher(lower) as $f = f_0 + v_r/\lambda_0 (= f_0 - v_r/\lambda_0)$. These two results may be combined into a single equation as follows.

$$f = \frac{v_0 \pm v_r}{v_0 \pm v_s} f_0 \tag{3}$$

  Note that the signs selected in such a way that the frequency will increase when either the source or receiver approaches each other.

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Experiment

Experimental equipment

- Ripple Tank
- Ripple Generator and Light Source
- Dipper
- Barrier

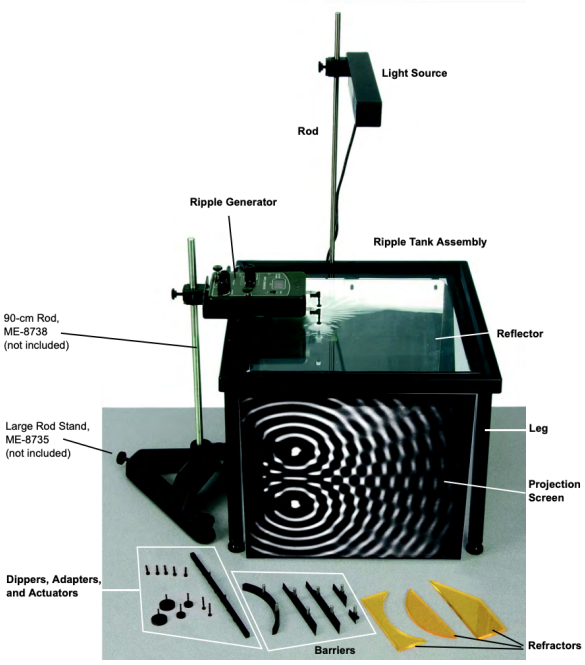


Figure 3: Ripple Tank Experiment Setup

Experimental Design

- **How to measure**

From this experiment, we have to measure various lengths(wavelength, distance from the source to point..) to get quantitative results. However it takes too much time to draw on paper, so we will use a tool "Capstone" from some experiments.

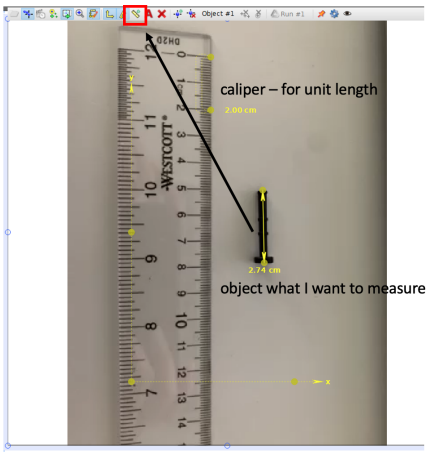


Figure 4: How to use capstone

First, go into 'video analysis' and insert a video. Set the unit length using the caliper(You have to take a picture of thing that you know the length well.), and press the button marked with a red box to measure the desired length or angle. Of course you can use any method for the measurement if you have more convenient or easier tool.

- **Basic Setup**

(a) Mount the light source onto its rod at the back edge of the ripple tank.

(b) Lock the water valve which is at the edge of the ripple tank assembly, and pour water in the assembly.

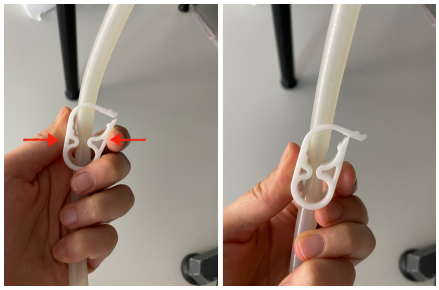
(c) Connect the plane wave dipper to the ripple arms. Adjust the ripple generator until the bottom of the plane wave dipper is barely in contact with the surface of the water.

(d) In each experiment, the depth of water, the amplitude and frequency of the wave and the height of the light source are different for the best results. Students have to control all variables to get a clear picture for the analysis.

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Warning

Make sure to lock the water valve before you pour water!!



Experimental Procedure

1. **Refraction**

   (a) Measure the height of the trapezoidal refractor to use as a caliper.

   (b) Arrange the trapezoidal refractor in the water in the middle of the tank so the rectangular end of the refractor is parallel to the plane wave dipper and about 8 cm from the dipper.

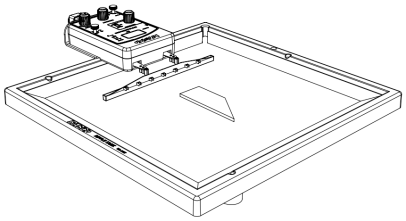


Figure 5: Refraction Experiment setup

   (c) Add just enough water to the tank so that **the refractor is evenly covered by less than 1 mm of water**.

   (d) Set the light source to **'STROBE'**. Set the ripple generator frequency to 15 Hz or less. Set the amplitude to slightly less than half of maximum and adjust it as necessary to make a clear pattern of plane waves.

   (e) Take a video and measure the wavelengths of deeper part and shallower part using capstone like Figure 6. For the more precise measurement, take at least three and average them.

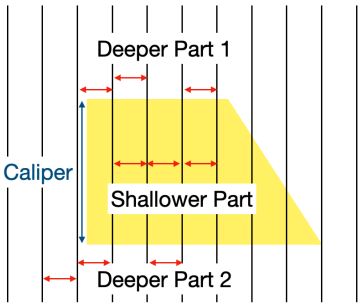


Figure 6: Measurement of the refraction experiment

2. **Double Slit Interference**

   (a) Place the short barrier of 5 cm between the two long barriers in the water to form two openings that are 1.5 cm long. Put the barriers parallel to the plane wave dipper as shown in Figure 7.

   (b) Set the barriers about 5 cm from the plane wave dipper.

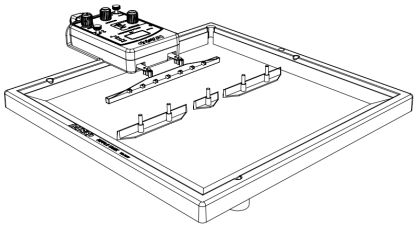


Figure 7: Interference Experiment setup

   (c) Set the frequency to slightly less than 20 Hz and the amplitude to slightly less than half of maximum.

   (d) Find and label the regions where the waves from the two slits tend to cancel each other or the rigions where the waves add together to make waves with higher peaks.

3. **Doppler Effect**

   (a) Put a standard dipper into one of the ripple arms of the ripple generator like Figure 8.

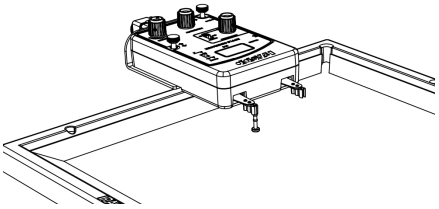


Figure 8: Doppler Effect Experiment setup

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(b) Start with the ripple generator's default frequency of f_0. Set the amplitude to about half of the maximum.

(c) Move the ripple generator at a constant speed.

(d) Using an image, measure the wavelengths at the front and back sides, respectively, of the moving ripple generator.

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Data Acquisition

Students must attach all pictures that were used for the analysis.

1. **Refraction**

   Note: To find λ, read the distance across many wavefronts and take its average.

No.	frequency (Hz)	wavelength, λ (cm)		
		Deeper part, 1	Deeper part, 2	Shallower part
1				
2				
3				
speed, v (cm/s)				

   Plot graphs of f versus λ^{-1} for deeper and shallower part, respectively. Find the wave speed in each part by fitting with $f = v\lambda^{-1} + b$.

Solution:

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Attach all pictures that were used for the analysis.

Solution:

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2. **Double Slit Interference**

Read the angles of maximum amplitude, or constructive interference, up to $m = \pm 2$. Alternatively, you may find the locations where nearby wave fronts cross each other.

frequency, f (Hz)		slit interval, d (cm)			λ (cm)	
m	-2	-1	0	1	2	Fitted d (cm)
θ (^{o})						

Plot $m\lambda$ versus $\sin\theta$, fit it with $m\lambda = d\sin\theta + e$ and write the fitted d in the table.

Solution:

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Attach all pictures that were used for the analysis.

Solution:

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3. **Doppler Effect** Find the source speed for each trial and write in the table.

Initial frequency, f_0 (Hz)	
Initial wavelength, λ_0 (cm)	
Initial wave speed, v_0 (cm/s)	

Trials		wavelength, λ (cm)	frequency, f (Hz)	source speed v_s (cm/s)
1	front			
	back			
2	front			
	back			

Explain in the box below how you obtained the source speed v_s.

Solution:

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Attach all pictures that were used for the analysis.

Solution:

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Analysis and Discussion

1. In Experiment 1, when the plane wave moves from the deeper water into the shallower, did the speed increase or decrease? Discuss the reason why the wave speed could change depending on the depth.

Solution:

2. In Experiment 2, does the plot look consistent with $m\lambda = d\sin\theta$? If not, explain why.

Solution:

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3. In Experiment 2, explain how you determined the angles of constructive interference.

Solution:

4. In Experiment 3, what will be the maximum v_s where Equation (3) can be used? What will happen if v_s exceeds that speed?

Solution:

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Summary and Conclusions

- Make a brief summary of the experimental results in 2 - 3 sentences. (In this experiment, what physical quantities did we measure and how did we measure them? If it includes any particularly important physical quantities, what are their values that we obtained?)

- Finally, make a short conclusion of the experiment. (What physics did we learn from this experiment? If it includes a law of physics, try to state it in a sentence instead of using an equation.)

Solution:

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